# CABLINE®-SS

High-data-rate transfer (16 Gbps/Lane), narrow depth, robust, max power delivery (AWG 34), 0.4 mm pitch, right angle vertical mating type micro-coaxial connector



**Product Specifications:**

| | | |
|---|---|---|
| Mating Type / Direction | | Vertical (Right Angle) |
| Board Pitch | | 0.4 mm |
| Contact Mating Length | | 0.3 mm |
| Mated Size | Height | 1.65 +/- 0.2 mm |
| | Width | 5.8 + (0.4 x pin count) mm |
| | Depth | 3 mm |
| Available Pin Count *1 | | 10, 14, 20, 30, 32, 35, 40, 50 |

**Applicable Cable Size:**

| | |
|---|---|
| Maximum O.D. | 0.4 mm |
| Micro-coaxial | AWG 36 *2, 38, 40, 42 |
| Twinaxial | - |
| Discrete | AWG 34, 36 |

**Applicable Standards (Reference Only):**

V-By-One US (16 Gbps/lane), HDMI 2.1 (12 Gbps/lane), USB 3.2 Gen 2 (10 Gbps/lane)

*1 Please inquire for pin counts not listed or outside of the pin count range.
*2 Provides option to use smaller AWG micro-coaxial cables with unmatched impedance that can be more suitable for power delivery applications in high density space constrained systems.

## High-data-rate transfer, ideal for 16 Gbps/lane applications

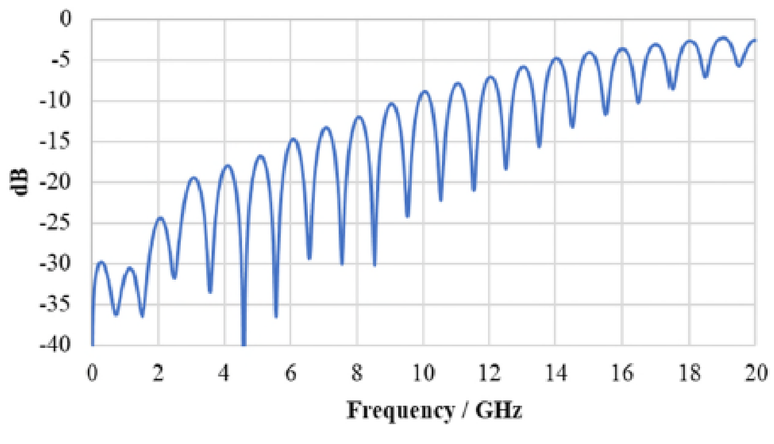
### Measurement results

Connector: CABLINE-SS 40 pin

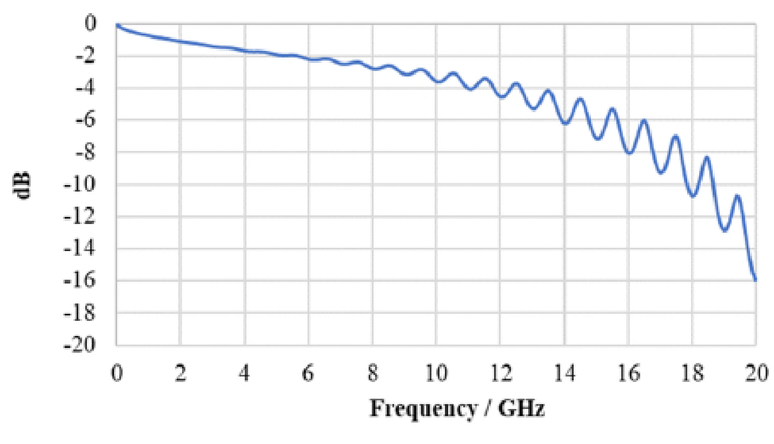
Cable: Micro-coaxial cable
AWG 40, 45 ohm , Length : 100 mm

Pin Assignment: GSSGSSG

Return Loss (SDD11)

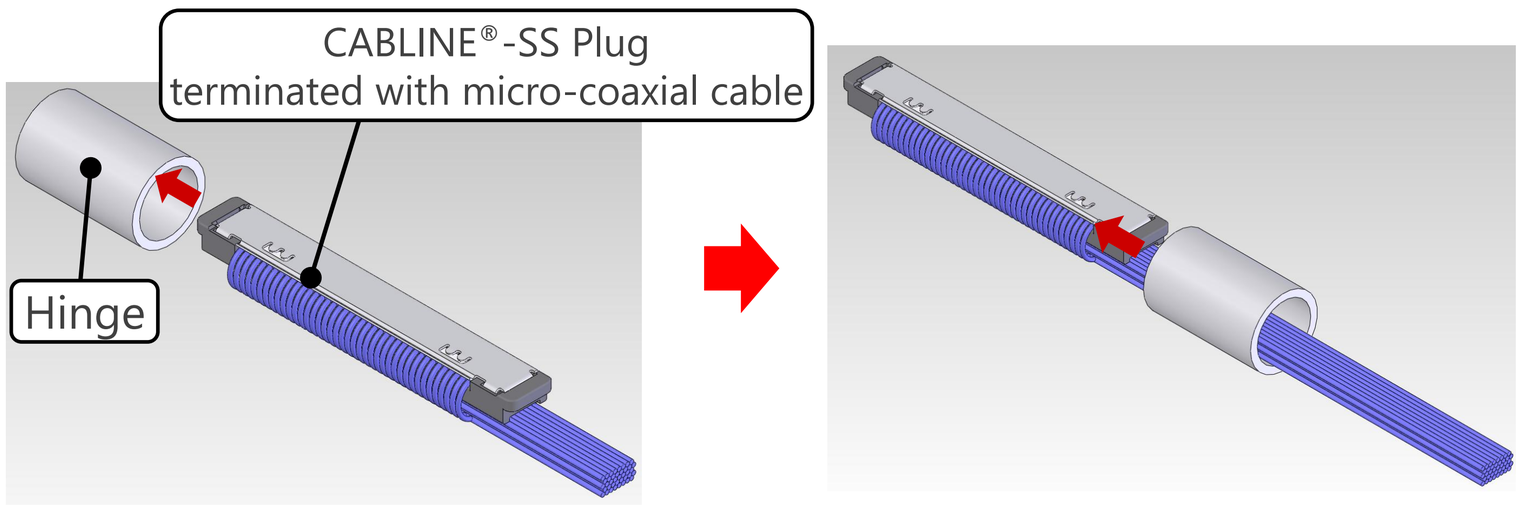


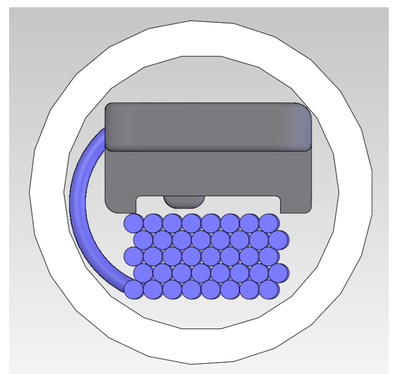

Insertion Loss (SDD21)




## Narrow depth design with maximum power delivery (AWG 34)

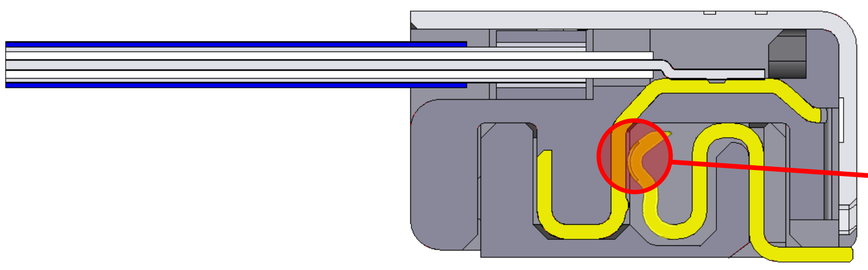
Narrow depth design enables go through small hinge after cable termination.



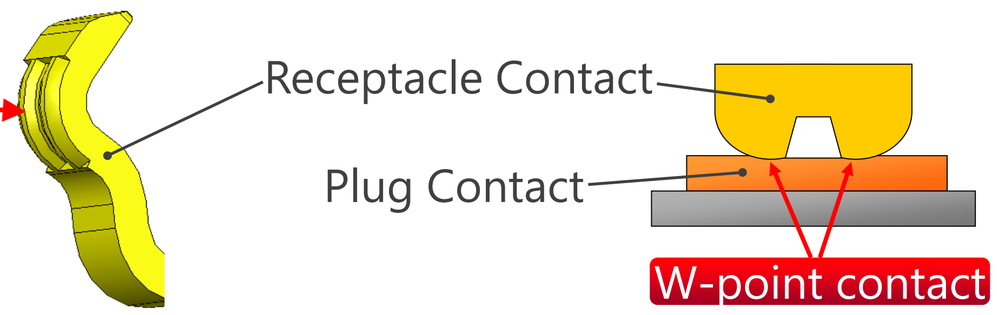



Side View

## Highly reliable contact using I-PEX's unique W-point design



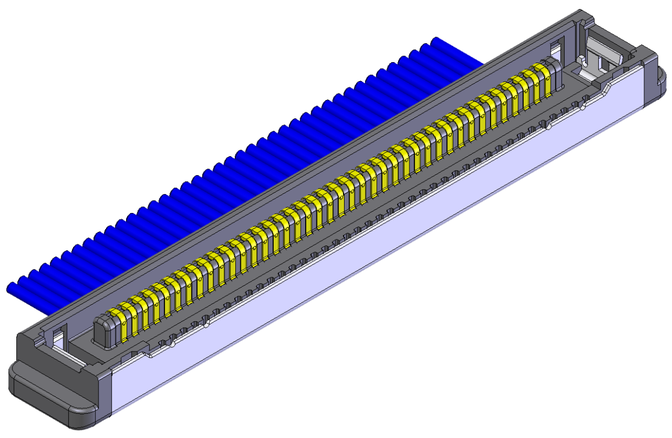
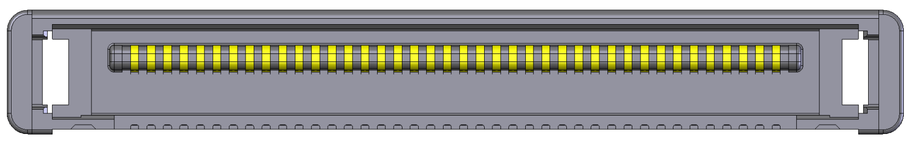
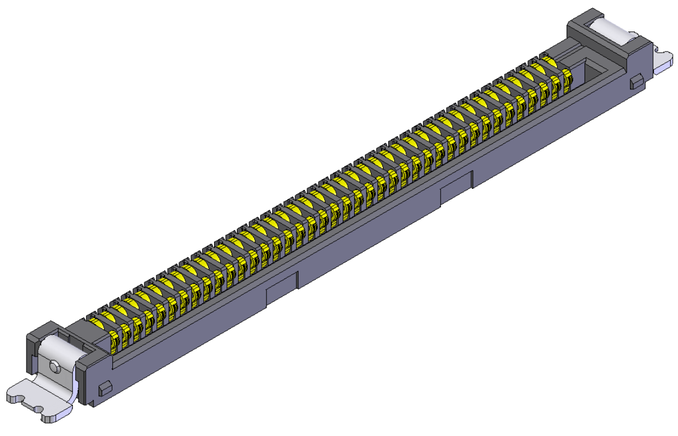
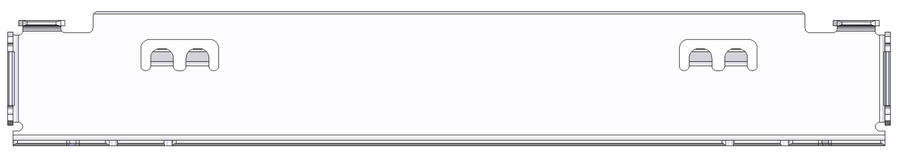
Plug & receptacle mated cross-section




I-PEX prevents poor contact with foreign particles between plug and receptacle contacts.
Provides better contact reliability than typical single-point contact designs.

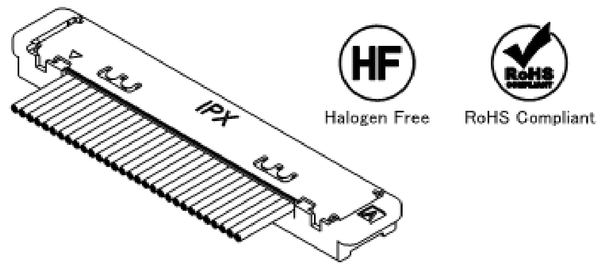
# Component Parts Details

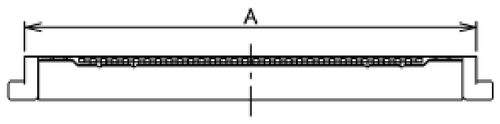
## Component Parts

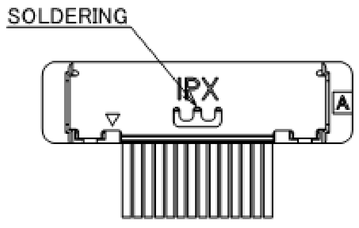
| Connector parts | Plug parts |
| --- | --- |
|  <br>PLUG |  <br>PLUG HOUSING ASSEMBLY |
|  <br>RECEPTACLE |  <br>PLUG METAL COVER |

# Plug for Cable Assembly

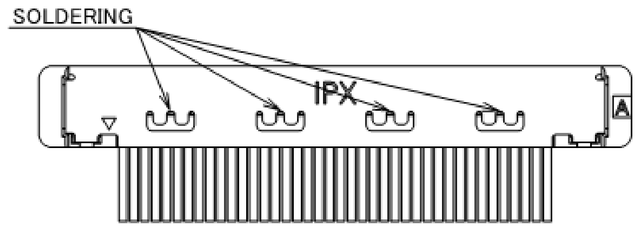
| Recommended P/N | 20380-R**T-06 (10P / 14P / 20P / 30P / 32P / 35P) | | | 20380-R**T-16 (40P / 50P) | |
|---|---|---|---|---|---|
| PART NO. | POS. | MATING RECEPTACLE P/N | A | B | C |
| 20380-R10T-06 | 10 | 20374-R10E-31 | 8.6 | 9.8 | 3.6 |
| 20380-R14T-06 | 14 | 20374-R14E-31 | 10.2 | 11.4 | 5.2 |
| 20380-R20T-06 | 20 | 20374-R20E-31 | 12.6 | 13.8 | 7.6 |
| 20380-R30T-06 | 30 | 20374-R30E-31 | 16.6 | 17.8 | 11.6 |
| 20380-R32T-06 | 32 | 20374-R32E-31 | 17.4 | 18.6 | 12.4 |
| 20380-R35T-06 | 35 | 20374-R35E-31 | 18.6 | 19.8 | 13.6 |
| 20380-R40T-16 | 40 | 20374-R40E-31 | 20.6 | 21.8 | 15.6 |
| 20380-R50T-16 | 50 | 20374-R50E-31 | 24.6 | 25.8 | 19.6 |



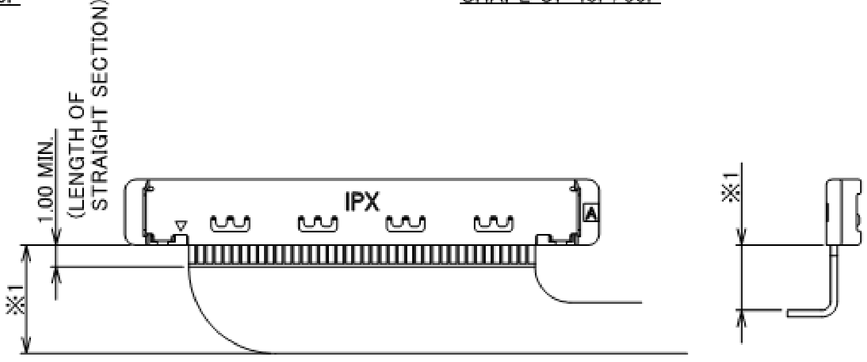




SHAPE OF 10P/14P/20P

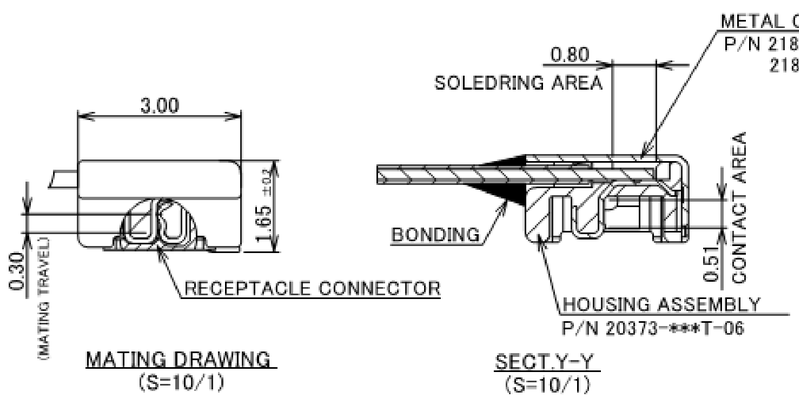



SHAPE OF 40P/50P




REFERENCE CABLE BENDING DIMENSIONS




MATING DRAWING
(S=10/1)

SECT.Y-Y
(S=10/1)

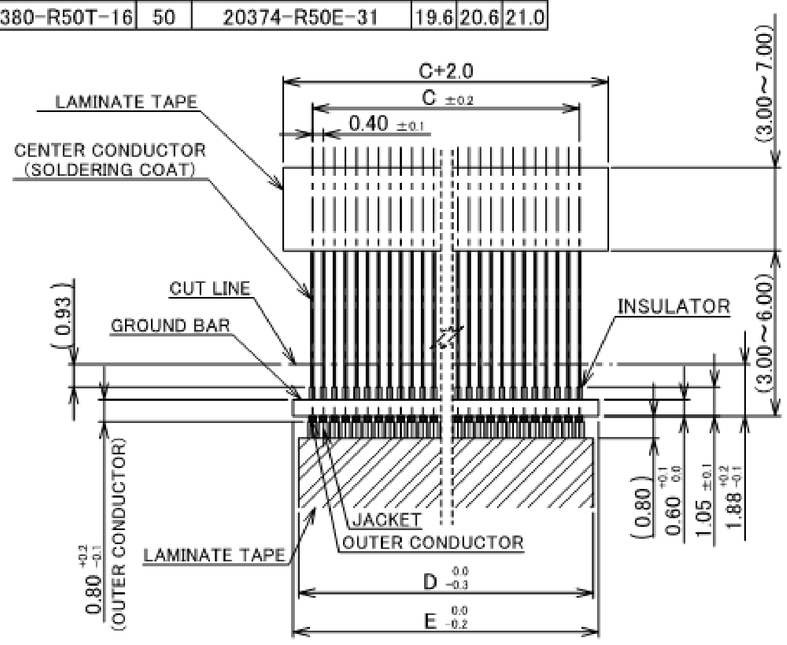
※1 : THE BEND DIMENSIONS MAY VARY BASED ON THE SIZE AND QUANTITY OF CABLES.
PLEASE CONFIRM THE MINIMUM BENDING RADIUS WITH I-PEX OR THE HARNESS MANUFACTURER.

| NO. | DISCRIPTION | MATERIAL | FINISH , REMARKS |
|---|---|---|---|
| 3 | METAL COVER | PHOSPHOR BRONZE | Sn REFLOW 1.00 μm MIN. |
| 2 | CONTACT | PHOSPHOR BRONZE | CONTACT AREA : Au 0.1 μm MIN. OVER Ni 1.00 μm MIN.<br>SOLDERING AREA : Au 0.1 μm MIN. OVER Ni 1.00 μm MIN. |
| 1 | HOUSING | LCP | UL94V-0, BLACK |

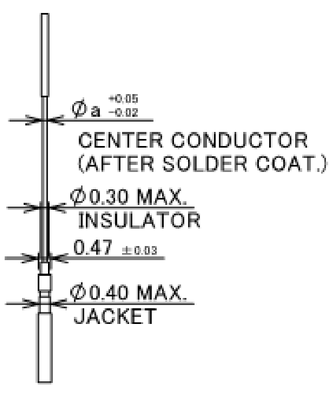
NOTES.
1. THE CABLE MUST NOT HOLD WHEN UN-MATING A CONNECTOR.

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| Recommended P/N | 20380-R**T-06 (10P / 14P / 20P / 30P / 32P / 35P) | | | 20380-R**T-16 (40P / 50P) | |
|---|---|---|---|---|---|
| PART NO. | POS. | MATING RECEP. P/N | C | D | E |
| 20380-R10T-06 | 10 | 20374-R10E-31 | 3.6 | 4.6 | 5.0 |
| 20380-R14T-06 | 14 | 20374-R14E-31 | 5.2 | 6.2 | 6.6 |
| 20380-R20T-06 | 20 | 20374-R20E-31 | 7.6 | 8.6 | 9.0 |
| 20380-R30T-06 | 30 | 20374-R30E-31 | 11.6 | 12.6 | 13.0 |
| 20380-R32T-06 | 32 | 20374-R32E-31 | 12.4 | 13.4 | 13.8 |
| 20380-R35T-06 | 35 | 20374-R35E-31 | 13.6 | 14.6 | 15.0 |
| 20380-R40T-16 | 40 | 20374-R40E-31 | 15.6 | 16.6 | 17.0 |
| 20380-R50T-16 | 50 | 20374-R50E-31 | 19.6 | 20.6 | 21.0 |




RECOMMENDED MICRO-COAXIAL CABLE DIMENSION (NTS)




MICRO-COAXIAL CABLE
(AWG #42,40,36)

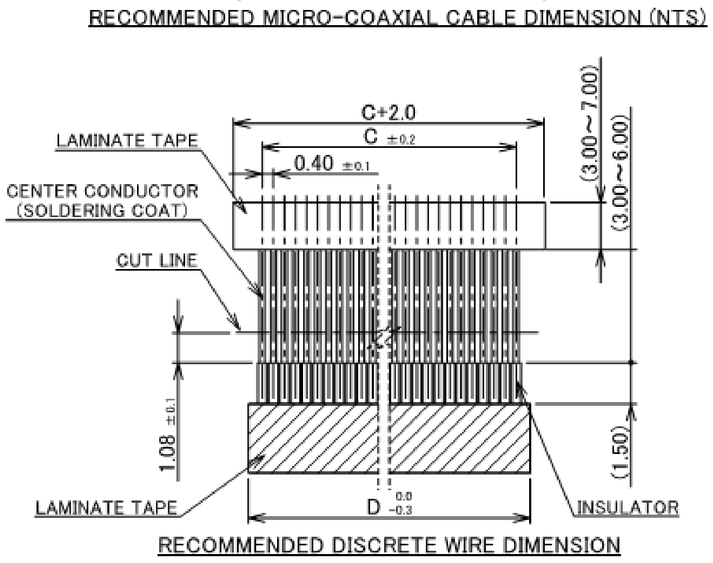
CHARACTERISTIC IMPEDANCE MACHING MICRO-COAXIAL CABLE

| | a |
|---|---|
| #38 | ϕ0.12 |
| #40 | ϕ0.09 |
| #42 | ϕ0.075 |

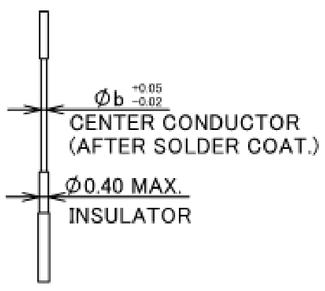
CHARACTERISTIC IMPEDANCE UN-MACHING MICRO-COAXIAL CABLE

| | a |
|---|---|
| #36 | ϕ0.15 |

MICRO-COAXIAL CABLE #36 : NOT RECOMMENDED FOR
HIGH SPEED SIGNAL TRANSFAER




RECOMMENDED DISCRETE WIRE DIMENSION




DISCRETE WIRE
(AWG #36,34)

DISCRETE WIRE DIMENSION

| | b |
|---|---|
| #34 | ϕ0.192 |
| #36 | ϕ0.15 |

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# Plug for Cable Assembly

| ITEMS | SPECIFICATION |
|---|---|
| APPLICABLE CABLE | MICRO-COAXIAL CABLE : AWG# 42 , 40 , 38 , 36<br>DISCRETE WIRE : AWG# 36 , 34 |
| RATING VOLTAGE | 100V AC (PER CONTACT PIN)<br>※THIS IS THE RATED VOLTAGE OF THE CONNECTOR.<br>PLEASE NOTE THAT THE RATED VOLTAGE MAY VARY IN THE HARNESS DEPENDING ON THE CABLES USED. |
| RATING AMPERAGE (FOR CONTACT) | 0.24A AC/DC [AWG#42](Per Contact Pin/Up to 50 Contacts)<br>0.25A AC/DC [AWG#40](Per Contact Pin/Up to 50 Contacts)<br>0.30A AC/DC [AWG#40](Per Contact Pin/Up to 32 Contacts)<br>0.33A AC/DC [AWG#38](Per Contact Pin/Up to 50 Contacts)<br>0.50A AC/DC [AWG#38](Per Contact Pin/Up to 12 Contacts)<br>0.35A AC/DC [AWG#36](Per Contact Pin/Up to 50 Contacts)<br>0.40A AC/DC [AWG#36](Per Contact Pin/Up to 37 Contacts)<br>0.80A AC/DC [AWG#36](Per Contact Pin/Up to 4 Contacts)<br>0.35A AC/DC [AWG#34](Per Contact Pin/Up to 50 Contacts)<br>1.00A AC/DC [AWG#34](Per Contact Pin/Up to 2 Contacts)<br>※TESTING BY A REAL MACHINE IS RECOMMENDED BECAUSE TEMPERATURE RISE MAY AFFECTED BY ACTUAL SITUATION. |
| OPERATING TEMPERATURE | 233～358K(-40℃～+85℃) |
| OPERATING HUMIDITY | 85% MAX.(NON-CONDENSING) |
| CONTACT RESISTANCE | INITIAL : 180mohm MAX. (AWG#34) / AFTER TEST : ⊿40mohm MAX.<br>275mohm MAX. (AWG#36)<br>360mohm MAX. (AWG#38)<br>600mohm MAX. (AWG#40)<br>700mohm MAX. (AWG#42)<br>(INITIAL CONTAINS THE CONDUCTOR RESISTANCE OF A CABLE 100mm) |
| GROUND SHELL RESISTANCE | INITIAL : 50mohm MAX. / AFTER TEST : ⊿40mohm MAX. |
| INSULATION RESISTANCE | INITIAL : 1000Mohm MIN. / AFTER TEST : 500Mohm MIN. |
| DIELECTRIC WITHSTANDING VOLTAGE | AC250V 1min |
| DURABILITY | 30 CYCLES |
| MATING FORCE (INITIAL / AFTER TEST) | 10P : 29.00N MAX. 32P : 33.40N MAX.<br>14P : 29.80N MAX. 35P : 34.00N MAX.<br>20P : 31.00N MAX. 40P : 35.00N MAX.<br>30P : 33.00N MAX. 50P : 38.00N MAX. |
| UNMATING FORCE (INITIAL / AFTER TEST) | 10P : INITIAL : 4.00N MIN. / AFTER 30 CYCLES : 2.87N MIN.<br>14P : INITIAL : 4.40N MIN. / AFTER 30 CYCLES : 3.23N MIN.<br>20P : INITIAL : 5.00N MIN. / AFTER 30 CYCLES : 3.76N MIN.<br>30P : INITIAL : 6.00N MIN. / AFTER 30 CYCLES : 4.65N MIN.<br>32P : INITIAL : 6.20N MIN. / AFTER 30 CYCLES : 4.84N MIN.<br>35P : INITIAL : 6.50N MIN. / AFTER 30 CYCLES : 5.07N MIN.<br>40P : INITIAL : 7.00N MIN. / AFTER 30 CYCLES : 5.50N MIN.<br>50P : INITIAL : 8.00N MIN. / AFTER 30 CYCLES : 6.41N MIN. |
| CABLE RETENTION FORCE | 10P : 4.90N MIN. 32P : 15.68N MIN.<br>14P : 6.86N MIN. 35P : 17.15N MIN.<br>20P : 9.80N MIN. 40P : 19.60N MIN.<br>30P : 14.70N MIN. 50P : 24.50N MIN. |
| PRODUCT SPECIFICATION | PRS-1239 |
| TEST REPORT | TR-04037 |
| INSTRUCTION MANUAL | HIM-04001 |
| ASSEMBLY MANUAL | ASM-04001 |
| APPEARANCE CRITERIA No. | QLS-A*** |

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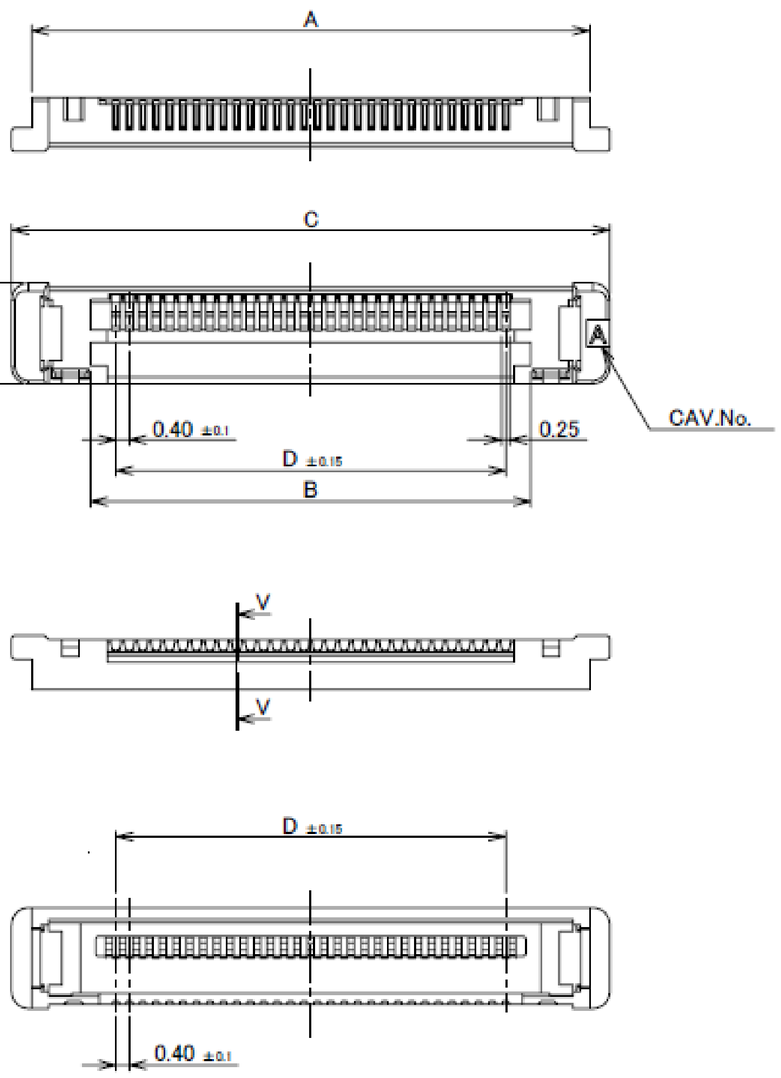
# Plug Housing Assembly

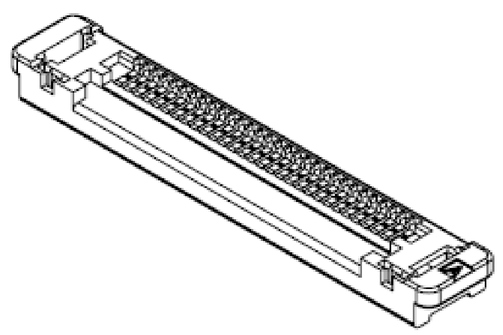
| Recommended P/N | | 20373-R**T-06 | | | |
|---|---|---|---|---|---|
| PART NO. | Pos. | A | B | C | D |
| 20373-R10T-06 | 10 | 8.60 | 5.10 | 9.80 | 3.60 |
| 20373-R14T-06 | 14 | 10.20 | 6.70 | 11.40 | 5.20 |
| 20373-R20T-06 | 20 | 12.60 | 9.10 | 13.80 | 7.60 |
| 20373-R30T-06 | 30 | 16.60 | 13.10 | 17.80 | 11.60 |
| 20373-R32T-06 | 32 | 17.40 | 13.90 | 18.60 | 12.40 |
| 20373-R35T-06 | 35 | 18.60 | 15.10 | 19.80 | 13.60 |
| 20373-R40T-06 | 40 | 20.60 | 17.10 | 21.80 | 15.60 |
| 20373-R50T-06 | 50 | 24.60 | 21.10 | 25.80 | 19.60 |

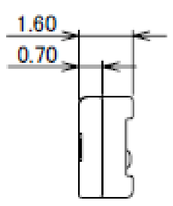
FOR AWG #34,#36,#40,#42

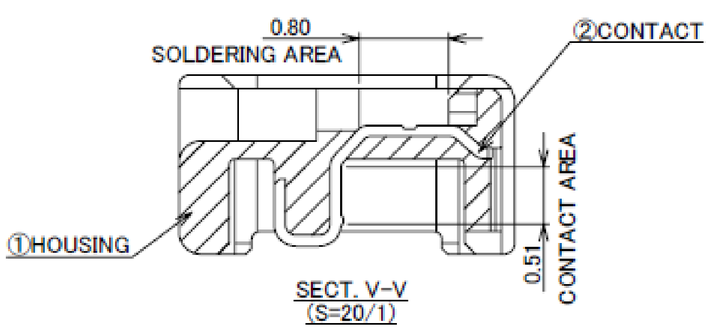
HF Halogen Free

RoHS Compliant






NOTES
1.THIS PART IS ASSEMBLED WITH METAL COVER (P/N 2182-0**-03(10P,14P,20P,30P,32P,35P) or 2182-0**-04(40P,50P)) AFTER SOLDERED CABLE, AND IT BECOMES P/N 20380-R**T-06(10P,14P,20P,30P,32P,35P) or 20380-R**T-16(40P,50P)

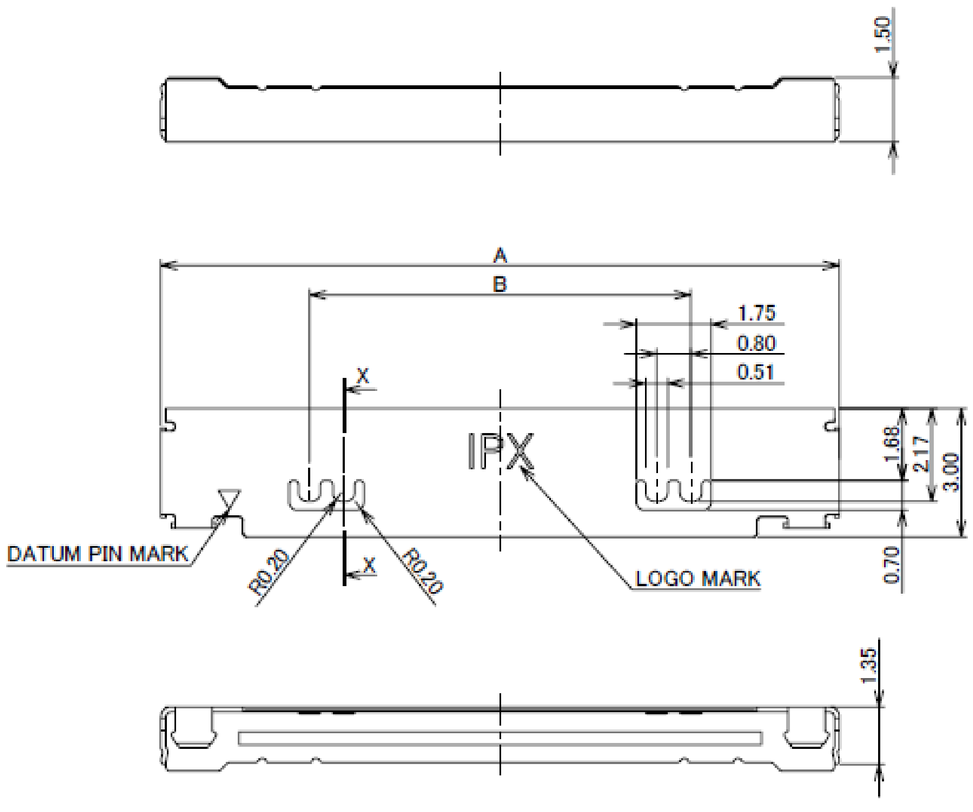
| NO. | DISCRIPTION | MATERIAL | FINISH , REMARKS |
|---|---|---|---|
| 2 | CONTACT | PHOSPHOR BRONZE | CONTACT AREA : Au 0.1 μm MIN. OVER Ni 1.00 μm MIN.<br>SOLDERING AREA : Au 0.1 μm MIN. OVER Ni 1.00 μm MIN. |
| 1 | HOUSING | LCP (RED PHOSPHORUS IS NOT USED) | UL94V-0, BLACK |

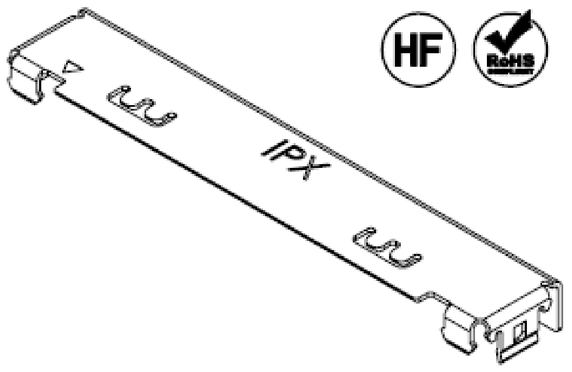
Rev.22

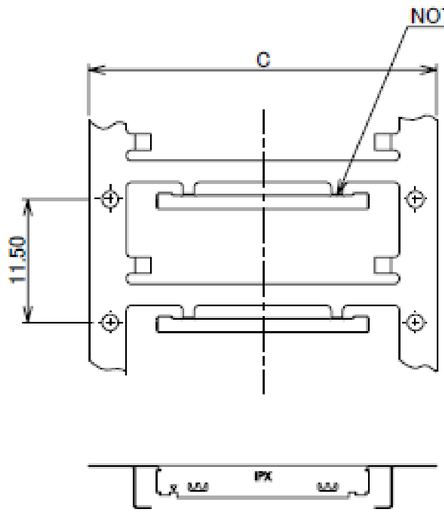
# Plug Metal Cover

| Recommended P/N | 2182-0**-03 (10P / 14P / 20P / 30P / 32P / 35P) | 2182-0**-04 (40P / 50P) |
|---|---|---|

| PART NO. | POS. | A | B | C |
|---|---|---|---|---|
| 2182-010-03 | 10 | 7.84 | – | 24.50 |
| 2182-014-03 | 14 | 9.44 | – | 24.50 |
| 2182-020-03 | 20 | 11.84 | – | 24.50 |
| 2182-030-03 | 30 | 15.84 | 8.90 | 28.50 |
| 2182-032-03 | 32 | 16.64 | 9.70 | 29.30 |
| 2182-035-03 | 35 | 17.84 | 10.90 | 30.50 |




WITH CAREER
(S=2/1)



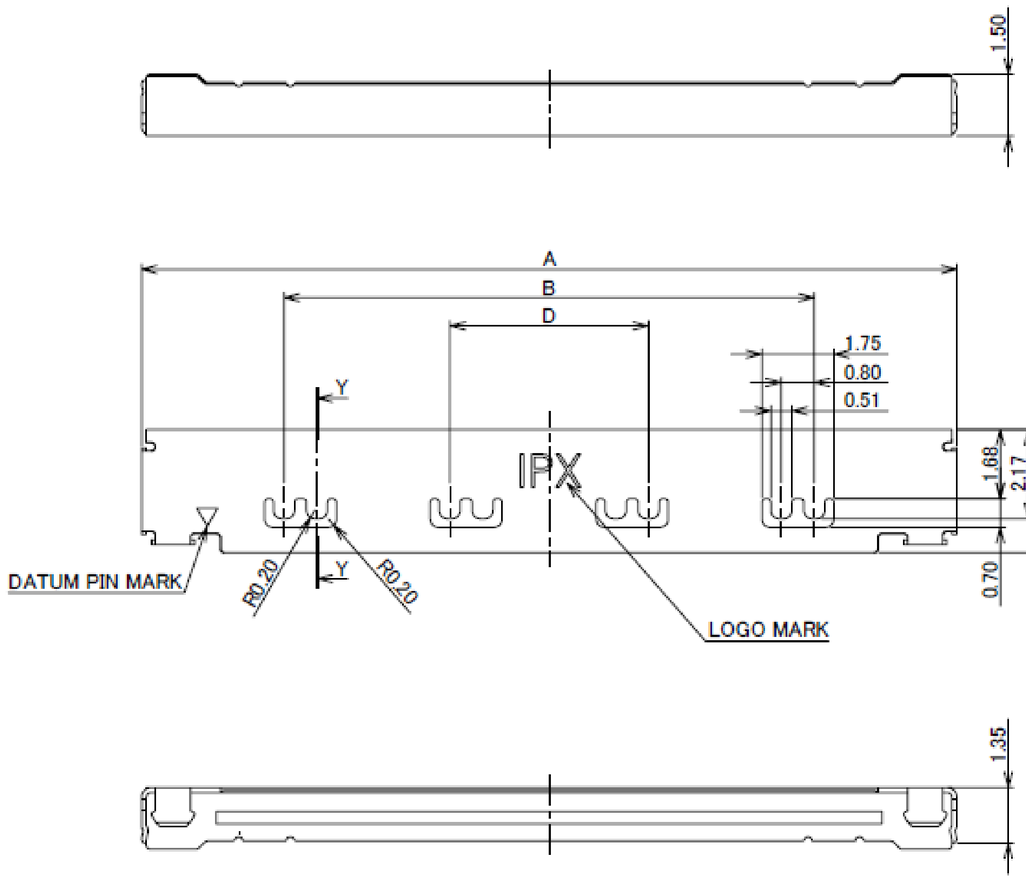

SHAPE OF 10P/14P/20P
(S=5/1)

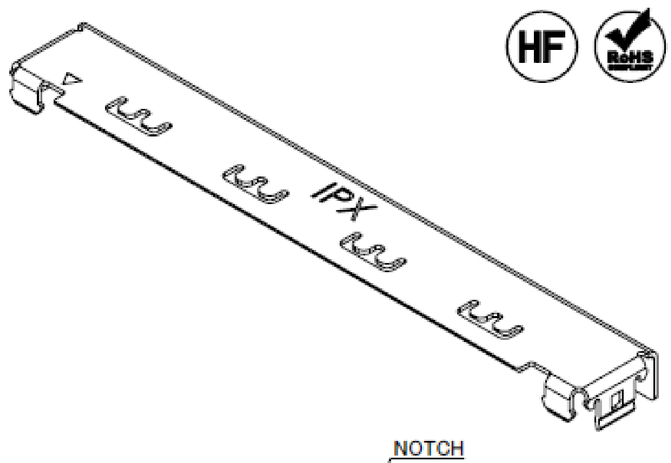
| NO. | DISCRIPTION | MATERIAL | FINISH , REMARKS |
|---|---|---|---|
| 1 | METAL COVER | PHOSPHOR BRONZE | Sn REFLOW 1.00 $\mu$ m MIN. |

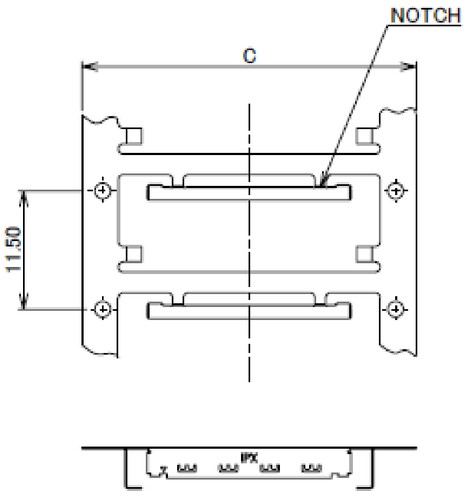
Rev.19

| Recommended P/N | 2182-0**-03 (10P / 14P / 20P / 30P / 32P / 35P) | 2182-0**-04 (40P / 50P) |
|---|---|---|

| PART NO. | POS. | A | B | C | D |
|---|---|---|---|---|---|
| 2182-040-04 | 40 | 19.84 | 12.90 | 32.50 | 4.84 |
| 2182-050-04 | 50 | 23.84 | 16.90 | 36.50 | 6.17 |




WITH CAREER
(S=2/1)

| NO. | DISCRIPTION | MATERIAL | FINISH , REMARKS |
|---|---|---|---|
| 1 | METAL COVER | PHOSPHOR BRONZE | Sn REFLOW 1.00 $\mu$ m MIN. |

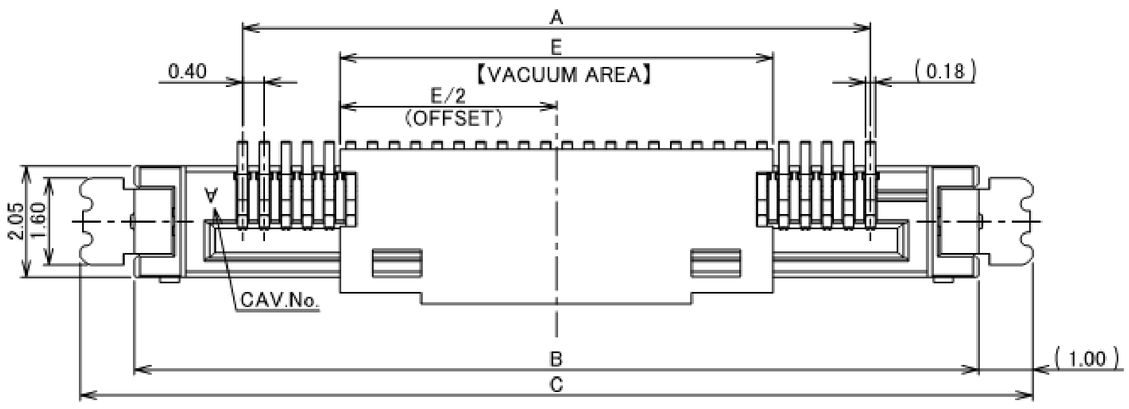
Rev.19

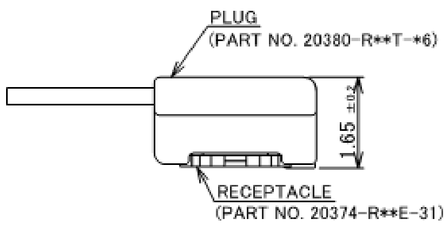
# Receptacle Assembly

| Recommended P/N | 20374-R**E-31 | | | | | |
|---|---|---|---|---|---|---|
| PART NO. | POS. | A | B | C | D | E |
| 20374-R10E-31 | 10 | 3.60 | 7.60 | 9.60 | 7.20 | 5.50 |
| 20374-R14E-31 | 14 | 5.20 | 9.20 | 11.20 | 8.80 | 5.50 |
| 20374-R20E-31 | 20 | 7.60 | 11.60 | 13.60 | 11.20 | 8.00 |
| 20374-R30E-31 | 30 | 11.60 | 15.60 | 17.60 | 15.20 | 8.00 |
| 20374-R32E-31 | 32 | 12.40 | 16.40 | 18.40 | 16.00 | 8.00 |
| 20374-R35E-31 | 35 | 13.60 | 17.60 | 19.60 | 17.20 | 8.00 |
| 20374-R40E-31 | 40 | 15.60 | 19.60 | 21.60 | 19.20 | 8.00 |
| 20374-R50E-31 | 50 | 19.60 | 23.60 | 25.60 | 23.20 | 8.00 |

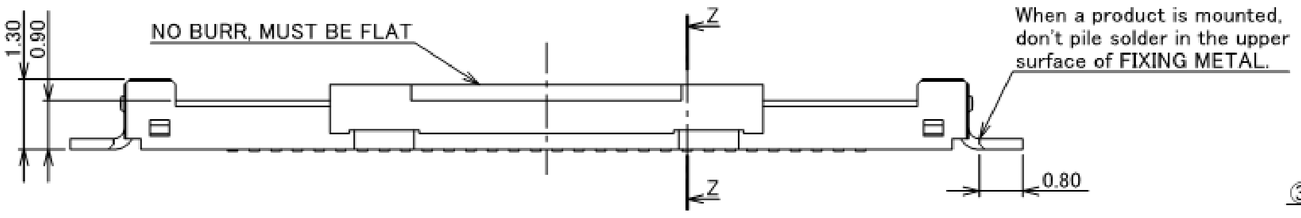
Halogen Free

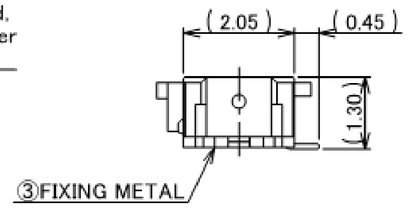
RoHS Compliant

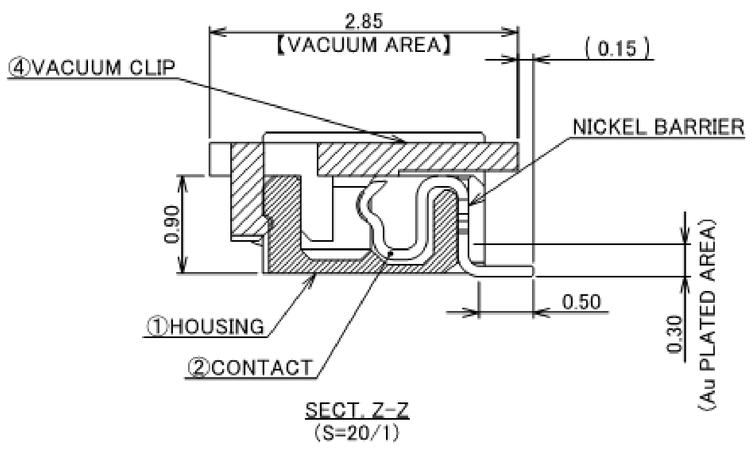



MATING CONDITION



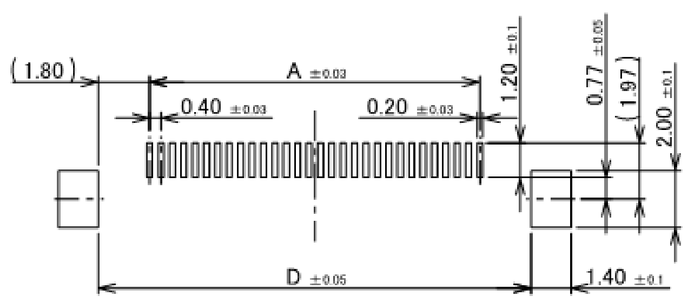



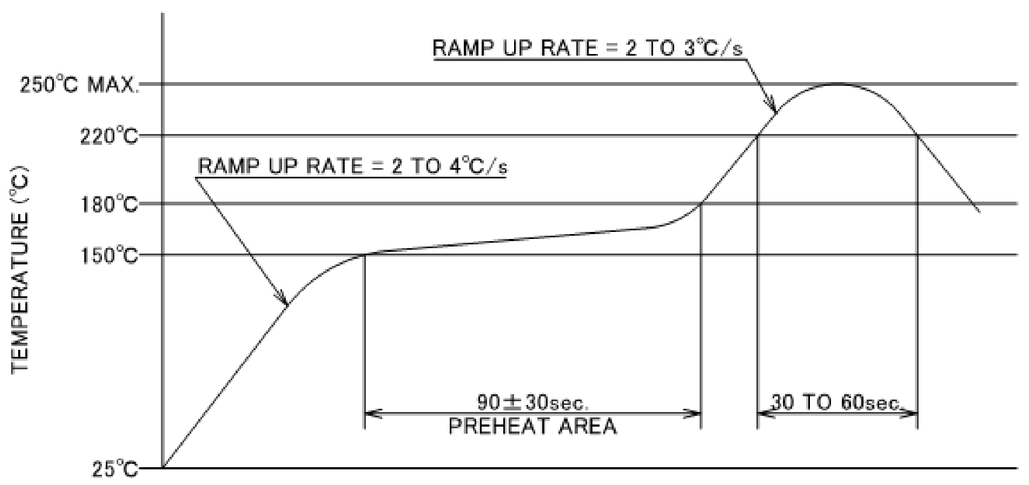

| NO. | DISCRIPTION | MATERIAL | FINISH , REMARKS |
|---|---|---|---|
| 4 | VACUUM CRIP | LCP | UL94V-0, BEIGE |
| 3 | FIXING METAL | PHOSPHOR BROZE | Sn/Cu(98/2) (LUSTER) 2.0μm MIN. OVER Ni 1.00μm MIN. |
| 2 | CONTACT | PHOSPHOR BROZE | CONTACT AREA : Au 0.1μm MIN. OVER Ni 1.00μm MIN.<br>SOLDERING AREA : Au 0.05μm MIN. OVER Ni 1.00μm MIN. |
| 1 | HOUSING | LCP(RED PHOSPHORUS IS NOT USED) | UL94V-0, BLACK |



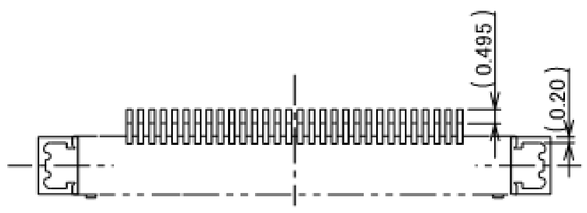
| PART NO. | POS. | A | D |
|---|---|---|---|
| 20374-R10E-31 | 10 | 3.60 | 7.20 |
| 20374-R14E-31 | 14 | 5.20 | 8.80 |
| 20374-R20E-31 | 20 | 7.60 | 11.20 |
| 20374-R30E-31 | 30 | 11.60 | 15.20 |
| 20374-R32E-31 | 32 | 12.40 | 16.00 |
| 20374-R35E-31 | 35 | 13.60 | 17.20 |
| 20374-R40E-31 | 40 | 15.60 | 19.20 |
| 20374-R50E-31 | 50 | 19.60 | 23.20 |




RECOMMENDED FOOTPRINT PATTERN (MOUNT SIDE)
Recommended Thickness of Metal Mask : t=0.13mm
Recommended Opening Ratio of Metal Mask : 79.6%




REFLOW TEMPERATURE PROFILE
SENJU METAL INDUSTRY CO., LTD. : M705-SHF(Sn96.5 Ag3.0 Cu0.5)




CONNECTOR ON RECOMMENDED FOOTPRIT PATTERN

Rev.33

# Receptacle Assembly

| ITEMS | SPECIFICATION |
|---|---|
| APPLICABLE CABLE | MICRO-COAXIAL CABLE : AWG# 42 , 40 , 38 , 36<br>DISCRETE WIRE : AWG# 36, 34 |
| RATING VOLTAGE | 100V AC (PER CONTACT PIN)<br>※THIS IS THE RATED VOLTAGE OF THE CONNECTOR.<br>PLEASE NOTE THAT THE RATED VOLTAGE MAY VARY IN THE HARNESS DEPENDING ON THE CABLES USED. |
| RATING AMPERAGE (FOR CONTACT) | 0.24A AC/DC [AWG#42](Per Contact Pin/Up to 50 Contacts)<br>0.25A AC/DC [AWG#40](Per Contact Pin/Up to 50 Contacts)<br>0.30A AC/DC [AWG#40](Per Contact Pin/Up to 32 Contacts)<br>0.33A AC/DC [AWG#38](Per Contact Pin/Up to 50 Contacts)<br>0.50A AC/DC [AWG#38](Per Contact Pin/Up to 12 Contacts)<br>0.35A AC/DC [AWG#36](Per Contact Pin/Up to 50 Contacts)<br>0.40A AC/DC [AWG#36](Per Contact Pin/Up to 37 Contacts)<br>0.80A AC/DC [AWG#36](Per Contact Pin/Up to 4 Contacts)<br>0.35A AC/DC [AWG#34](Per Contact Pin/Up to 50 Contacts)<br>1.00A AC/DC [AWG#34](Per Contact Pin/Up to 2 Contacts)<br>※TESTING BY A REAL MACHINE IS RECOMMENDED BECAUSE TEMPERATURE RISE MAY AFFECTED BY ACTUAL SITUATION. |
| OPERATING TEMPERATURE | 233～358K (-40℃～85℃) |
| OPERATING HUMIDITY | 85% MAX. (NON-CONDENSING) |
| CONTACT RESISTANCE | INITIAL : 180mohm MAX. (AWG#34) / AFTER TEST : ⊿40mohm MAX.<br>275mohm MAX. (AWG#36)<br>360mohm MAX. (AWG#38)<br>600mohm MAX. (AWG#40)<br>700mohm MAX. (AWG#42)<br>(INITIAL CONTAINS THE CONDUCTOR RESISTANCE OF A CABLE 100mm) |
| GROUND SHELL RESISTANCE | INITIAL : 50mohm MAX. / AFTER TEST : ⊿40mohm MAX. |
| INSULATION RESISTANCE | INITIAL : 1000Mohm MIN. / AFTER TEST : 500Mohm MIN. |
| DIELECTRIC WITHSTANDING VOLTAGE | AC250V 1min |
| DURABILITY | 30 CYCLES |
| MATING FORCE (INITIAL / AFTER 30 CYCLES) | 10P : 29.00N MAX. 32P : 33.40N MAX.<br>14P : 29.80N MAX. 35P : 34.00N MAX.<br>20P : 31.00N MAX. 40P : 35.00N MAX.<br>30P : 33.00N MAX. 50P : 38.00N MAX. |
| UNMATING FORCE | 10P : INITIAL : 4.00N MIN. / AFTER 30 CYCLES : 2.87N MIN.<br>14P : INITIAL : 4.40N MIN. / AFTER 30 CYCLES : 3.23N MIN.<br>20P : INITIAL : 5.00N MIN. / AFTER 30 CYCLES : 3.76N MIN.<br>30P : INITIAL : 6.00N MIN. / AFTER 30 CYCLES : 4.65N MIN.<br>32P : INITIAL : 6.20N MIN. / AFTER 30 CYCLES : 4.84N MIN.<br>35P : INITIAL : 6.50N MIN. / AFTER 30 CYCLES : 5.07N MIN.<br>40P : INITIAL : 7.00N MIN. / AFTER 30 CYCLES : 5.50N MIN.<br>50P : INITIAL : 8.00N MIN. / AFTER 30 CYCLES : 6.41N MIN. |
| COPLANARITY | 0.10 MAX. |
| PRODUCT SPECIFICATION | PRS-1239 |
| TEST REPORT | TR-04037 |
| PACKING STANDARD | 300-359 |
| INSTRUCTION MANUAL | HIM-04001 |
| APPEARANCE CRITERIA No. | QLS-A*** |

Rev.33

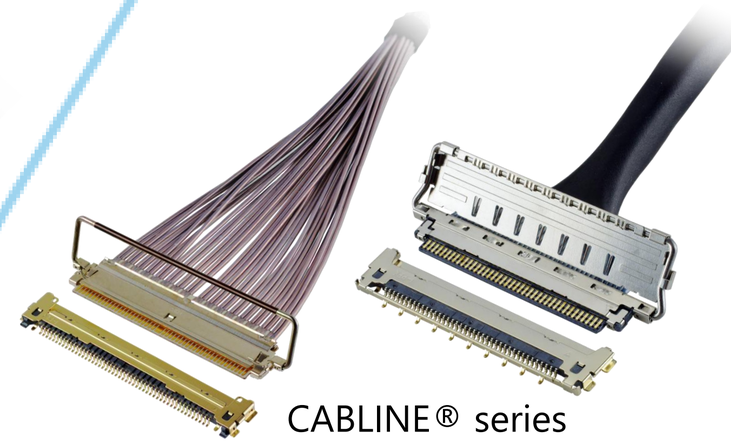
# Custom Connectors Available

## RF Connectors

MHF® series



## Micro-Coaxial/Twinaxial Connectors



CABLINE® series

## Wire-to-Board Connectors/ Terminals

AP series

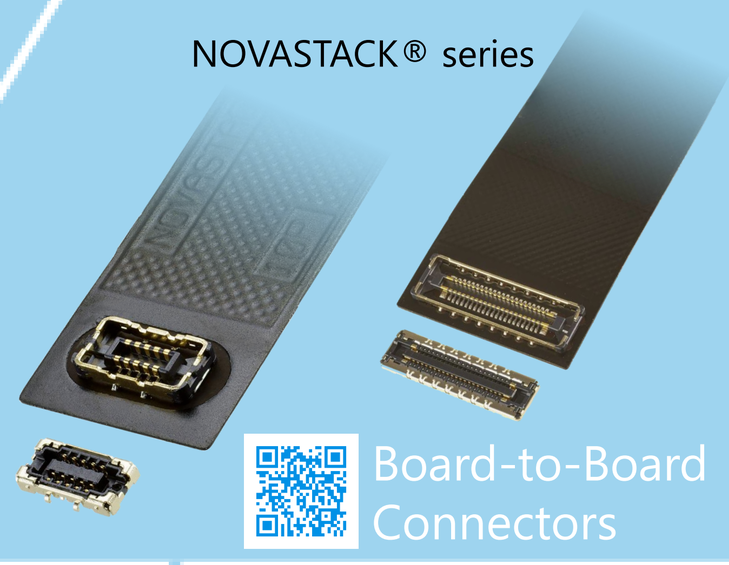


ISH® series

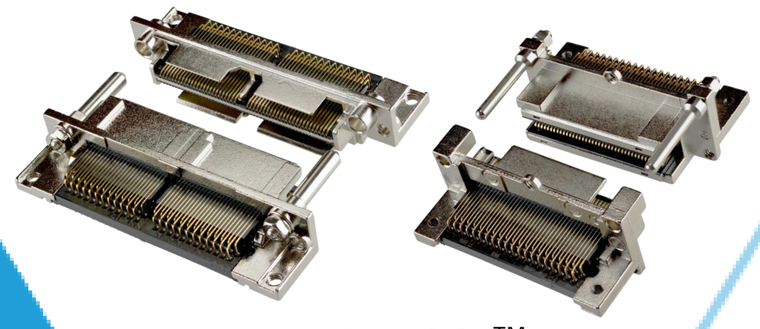


## Board-to-Board Connectors

NOVASTACK® series



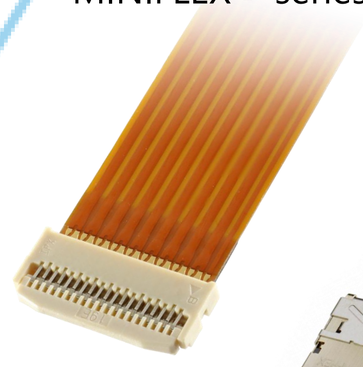
## I/O Connectors



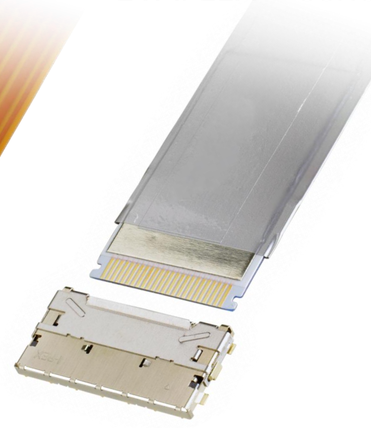
MINIDOCK™ series

## FPC/FFC Connectors

MINIFLEX® series



EVAFLEX® series



## Inquiry

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